

Objectives

To research and justify the selection of appropriate machine learning algorithms for predicting student dropout using the OULAD dataset. This involved reviewing algorithm characteristics against the requirements of the prediction task and recommending a final set of models for implementation by the team.

Task Completed

Algorithm Selection Rationale

The student dropout prediction problem is a binary classification task (is_dropout: 0 = non-dropout, 1 = dropout) with approximately 32,000 records and a mix of numerical and categorical features. Based on these requirements, three algorithms were selected:

1. Logistic Regression

Logistic Regression was selected as the baseline model. It is well-suited for binary classification and produces interpretable probability outputs, which are useful for understanding the relative contribution of each feature to the dropout risk. Its simplicity also makes it an important benchmark, if more complex models cannot significantly outperform it, there may be limited justification for their added complexity.

- Type: Linear, parametric classifier.
- Strengths: Highly interpretable, computationally efficient, outputs calibrated probabilities.
- Limitation: Assumes linear decision boundary; may underfit complex non-linear patterns in student engagement data.

2. Random Forest

Random Forest was selected as a non-linear ensemble model. Student dropout behavior is influenced by many interacting factors (engagement, prior attempts, demographics), and Random Forest is well-suited to capture these interactions without requiring manual feature engineering. Its bagging mechanism also reduces variance, making it robust to noise in the dataset.

- Type: Ensemble, tree-based (bagging).
- Strengths: Handles non-linearity and feature interactions; robust to noisy/irrelevant features; provides built-in feature importance scores.
- Limitation: Less interpretable than Logistic Regression; slower at inference time.

3. XGBoost (Extreme Gradient Boosting)

XGBoost was selected as the primary high-performance model. It is a gradient boosting framework that builds trees sequentially to correct the errors of previous trees, making it highly effective on

structured tabular data. It includes built-in regularization (L1 and L2) to prevent overfitting, and it handles class imbalance through the `scale_pos_weight` parameter. XGBoost has consistently demonstrated state-of-the-art performance in academic and industry classification benchmarks.

- Type: Ensemble, tree-based (gradient boosting).
- Strengths: High predictive accuracy on tabular data; handles missing values natively; supports regularization; fast and scalable via parallel processing.
- Limitation: More hyperparameters to tune compared to LR or RF; higher computational cost.

Algorithm Comparison Summary

The table below summarizes the rationale for each selected algorithm:

Algorithm	Strengths	Limitations	Selected?
Logistic Regression	Simple linear model; interpretable coefficients; fast to train; good baseline for binary classification.	May underfit non-linear patterns in student behavior data.	Yes
Random Forest	Handles non-linearity; robust to noisy/irrelevant features; built-in feature importance; low risk of overfitting via bagging.	Less interpretable than LR; slower inference.	Yes
XGBoost	State-of-the-art on tabular data; gradient boosting handles class imbalance better; regularization prevents overfitting; supports feature importance.	More hyperparameters to tune; higher compute cost.	Yes

Justification for Multi-Algorithm Approach

Rather than committing to a single algorithm upfront, three models were selected to allow for a comparative evaluation. This approach is standard practice in machine learning projects because:

- It allows the team to verify whether complex models (XGBoost, RF) provide a significant improvement over a simpler baseline (Logistic Regression).
 - Different algorithms may capture different patterns in the data — comparing them reveals which type of relationship (linear vs. non-linear) is most predictive.
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- Model selection based on empirical test results is more defensible than selecting a model based on theoretical properties alone.

All three models are compatible with the preprocessed feature set (StandardScaler-normalized numerical features and label-encoded categorical features) and support scikit-learn's unified fit/predict API, making parallel implementation straightforward.

Results and Analysis

The algorithm selection phase produced a clear and justified set of three models for the team to implement. The chosen algorithms cover the full spectrum from interpretable linear models to high-performance ensemble methods, enabling a rigorous comparative evaluation. The decision to include XGBoost was particularly well-supported given its proven track record on structured educational data in related studies.

Post-implementation results confirmed the selection rationale: XGBoost achieved the highest accuracy (84.7% after tuning) and ROC-AUC (0.9202), followed closely by Random Forest (0.9146), with Logistic Regression providing a solid baseline (0.8892). This outcome validates the multi-algorithm approach and the prioritization of gradient boosting for further optimization.

Challenges and Solutions

Challenge	Solution
Deciding whether to include deep learning models (e.g., neural networks) alongside classical ML algorithms.	After reviewing the dataset size (~32,000 records) and the structured tabular nature of the features, classical ML algorithms were deemed more appropriate. Deep learning models typically require larger datasets and offer limited interpretability advantages for this type of structured data. The chosen algorithms are better suited for educational analytics contexts where model explainability is important for stakeholder trust.

Conclusion

The algorithm research and selection task identified three well-suited classification models — Logistic Regression, Random Forest, and XGBoost,for predicting student dropout using the OULAD dataset. The selection was grounded in the characteristics of the prediction task: binary classification, tabular structured data, moderate dataset size, and the need for both high accuracy and interpretability.

The multi-algorithm approach enabled the team to empirically compare model performance rather than relying on theoretical assumptions. XGBoost emerged as the recommended primary model based on post-implementation evaluation results, confirming the rationale established during the selection phase.